

Wenqin Chen

📍 Seattle, WA | ✉ wenqinchenphy@gmail.com | ☎ (206) 535-5195 | 🌐 wen-qin-chen | 🐙 wenqinchen98

EDUCATION

- University of Washington**, Seattle, WA 2021 – 2027
PhD in Physics. Advisors: Di Xiao, Ting Cao
- Thesis: Deep Learning and Quantum Many-Body Theory for Condensed Matter Physics
- Xi'an Jiaotong University**, Xi'an, China 2016 - 2021
BS in Physics (Honors)
- University of California**, Berkeley, CA 2019
Exchange student in Physics

EXPERIENCE

- Los Alamos National Laboratory (T-4/CNLS)**, Los Alamos, NM Summers 2023 & 2024
Research Intern. Mentor: Shi-Zeng Lin
- Built an end-to-end self-consistent mean-field (SCF) numerical framework to search for skyrmion crystals and quantum anomalous Hall order in strongly correlated many-body electronic systems.
 - Ran a large computational campaign of parameter scans, scaled on the Department of Energy's NERSC Perlmutter supercomputer (SLURM, 128 cores/task); open-sourced a reusable ~28k-line, 148-test research codebase on GitHub.
 - Separately, delivered two first-author papers on phonon magnetism in quantum materials [PRB 2025] [arxiv 2025] in two consecutive summers, with predictions guiding the lab's experiments toward a co-authored manuscript (in preparation).
- Lawrence Berkeley National Laboratory**, Berkeley, CA 2019
Research Intern. Mentor: Joseph Orenstein
- Developed numerical simulations to interpret optical birefringence experimental data, co-authored in [Nat. Mater. 2020].

PROJECTS

- Attention-Based Neural Networks for Discovering Topological Quantum Materials**
- Transformer neural-network quantum state (NQS) in JAX for many-body electronic systems, sampled with variational Monte Carlo, optimized by natural-gradient method (stochastic reconfiguration), pretrained with physics-informed SCF solutions, and benchmarked against exact-diagonalization ground truth.
 - NQS training on UW Tillicum H200 GPU cluster and Azure cloud compute.
 - Open-sourced a reusable, pytest-tested NQS module (NaN-robust gradient clipping, learning-rate schedules, Hartree-Fock warm starts). Paper *Quantum Anomalous Hall Crystal* in preparation.
- Fault-Tolerant Topological Quantum Computing with Majorana Zero Modes**
- Ran large-scale, time-dependent numerical simulations of non-Abelian Majorana braiding in a Tetron device, made robust to an Andreev bound state (the chief obstacle to Majorana qubits) via geometric-phase cancellation; first author [PRB 2022].
 - Co-developed a minimal quantum-dot protocol to braid Majoranas with electrical fermion-parity readout [SCPMA 2021].
 - First-authored a quantum-geometric theory of Cooper pairs in Majorana-hosting chiral superconductors [arxiv 2026].

PUBLICATIONS

8 papers including *Nature Materials* and *Physical Review Letters* · 180+ citations · h-index 5 · full list on 📖 Google Scholar

TECHNICAL SKILLS

- Machine learning:** Neural networks (transformer/attention architectures, neural quantum states), variational Monte Carlo, natural-gradient optimization (stochastic reconfiguration), training stabilization (gradient clipping, learning-rate schedules, warm starts)
- Programming:** Python (NumPy, SciPy, pandas), JAX (Flax, Optax), NetKet, Bash/Linux, Git, pytest
- Scientific computing:** Electronic-structure methods (exact diagonalization, self-consistent-field Hartree-Fock), Bogoliubov-de Gennes / Nambu modeling of superconductor-semiconductor (Majorana) nanowire devices, Berry-curvature and transport (Kubo) calculations, stochastic methods, numerical linear algebra
- Infrastructure:** GPU training in JAX (H200), SLURM/HPC job orchestration, Azure cloud compute, reproducible research pipelines

AWARDS AND LEADERSHIP

- Clean Energy Institute & Thouless Quantum Matter Graduate Fellowships (UW); Outstanding Undergraduate Thesis (XJTU)